Course Title: Artificial Intelligence
Term: Spring 2025
Quiz Title: Quiz 2 – CSP & Regression
Time: 60 minutes

Question 1: Mathematical Inference from Constraints and AC-3 in CSP

Problem:

A CSP problem is defined with 4 variables A, B, C, D and the following domains:

- $A \in \{1, 2, 3\}$
- $B \in \{2, 3\}$
- $C \in \{1, 3\}$
- $D \in \{1, 2, 3\}$

The following binary constraints exist between the variables:

- $A \neq B$
- $B < C$
- $C \neq D$
- $A + D \leq 4$

Tasks:

a) List all arcs that initially enter the queue in the AC-3 algorithm.

b) Step-by-step, execute the AC-3 algorithm. Present in a table including:

- Selected arc
- Domain changes
- Whether new arcs are added or not

c) If, after executing AC-3, the domain of variable B becomes empty, can we conclude anything about the satisfiability of the problem? Provide a complete mathematical analysis.

Question 2: Contour plots

(a) Consider the following cost function:

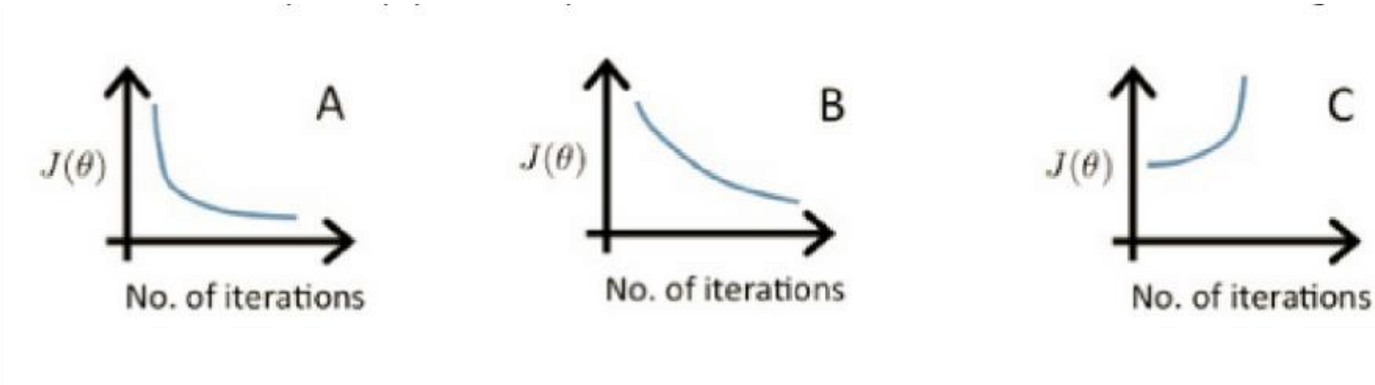
$$J(w)=5w_1^2+50w_2^2$$

Does convergence have equal speed in both directions? Why?

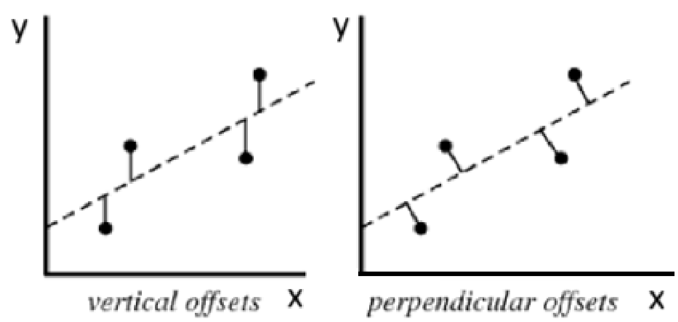
(b) Between SGD, GD and minibatch-GD which algorithm is more likely to exhibit high variance in the optimization path? Why?

(d) Draw the trajectory of SGD, GD, and Mini-batch GD algorithms on an elliptical contour plot and explain the reasons for the differences in their paths.

(e) compare learning rate value in these three images.



(f) In the linear regression problem, which of the following distances do you use?



(g) In a regression problem with a large number of features, which of the methods, Ridge or Lasso, provides a simpler model? Why?

Question 3: Advanced Consistency Concepts: K-Consistency and Its Applications in CSPs

Introduction: Beyond Arc Consistency (2-Consistency), higher levels of consistency (such as K-Consistency) exist in CSPs, which can further reduce the search space and aid in understanding the problem structure.

a) Analysis of K-Consistency and Path Consistency:

- Formally define the general concept of K-Consistency for an arbitrary integer $K \geq 2$. Then, explain why Arc Consistency (2-Consistency) and Path Consistency (3-Consistency) are specific cases and examples of K-Consistency.
- Consider the "Map Coloring" problem. Provide a small, specific example of this problem with at least 4 regions and 3 colors (draw the regions and their connections). Then, conceptually and step-by-step, explain how enforcing Path Consistency (3-Consistency) can eliminate more values from variable domains compared to just Arc Consistency (2-Consistency). Describe the key differences in how constraints are checked for Path Consistency compared to Arc Consistency.

b) Application of Consistency in Non-binary Constraint Problems and Practical Analysis:

- Explain how a non-binary constraint of degree 3 or higher (e.g., $C(X_1, X_2, X_3)$ involving three variables) can be transformed into a set of binary constraints, allowing the application of Arc Consistency algorithms. Describe the "Dual Graph Transformation" approach and illustrate how, through this transformation, applying Arc Consistency on the "transformed problem" can be equivalent to enforcing a higher level of consistency on the "original problem" with non-binary constraints.
- Consider a class scheduling problem, which includes complex constraints such as "Class A and Class B must be scheduled on the same day, but Class C must be on a different day, and all three classes must be taught by the same professor." Given the concepts of Arc Consistency and Path Consistency, analyze how applying these levels of consistency can practically contribute to a more efficient solution of this problem. Specifically, explain what types of variable domains and constraint structures in this problem would benefit most from the application of Arc Consistency and/or Path Consistency, and why.

c) Importance of Variable and Value Ordering in Backtracking Search:

- Explain how the strategic use of heuristics such as Minimum Remaining Values (MRV) and Least Constraining Value (LCV) within a backtracking search algorithm, which is executed after consistency enforcement (e.g., AC-3), can significantly reduce the search space and improve the overall performance of the CSP algorithm. Provide a brief example (e.g., a partial state in a small CSP) demonstrating how incorrect variable or value choices (not following MRV and LCV) can lead to more "Backtracking" and slower solution finding.

Question 4

You are given the following training dataset:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 4 \end{bmatrix}$$

Each row in matrix X represents a training sample with a bias term (first column is bias).

Update the weights for two steps using batch gradient descent algorithm. (α=0.01 and initialize the weights with zero.)